GAREL'S TRACE MAP ON THE DIFFERENTIAL FORMS OF A FINITE SYNTOMIC MORPHISM, CONSTRUCTED AND SHOWN UNIQUE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file derham.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-derham/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The upstairs Garel map from the differentials of the source to the pullback of the differentials of the target twisted by the determinant of the cotangent complex. Same chapter derham.tex, label lemma-Garel-upstairs (invoked at proof line 4062). It supplies the canonical map $c^p_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \to f^*\Omega^p_{X/\mathbf{Z}} \otimes \det(\NL_{Y/X})$ used in the locally Noetherian case.
- The Tate map identifying the determinant of the cotangent complex with the dualizing sheaf, and the trace of a finite morphism. Discriminants, label discriminant-proposition-tate-map (invoked at proof line 4068), Section \ref{discriminant-section-discriminant} (invoked at proof line 4055) and Section \ref{discriminant-section-finite-morphisms} (invoked at proof lines 4080 and 4092). They give the isomorphism $c_{Y/X} : \det(\NL_{Y/X}) \to \omega_{Y/X}$ sending δ to τ, and the evaluation $f_*\omega_{Y/X} \to \mathcal{O}_X$ sending $\tau_{Y/X}$ to $\operatorname{Trace}_f(1)$, which is what turns the twisted map into the map of the lemma and gives property (1).
- The local universal finite syntomic model, and its smooth and etale members. Discriminants, labels discriminant-lemma-locally-comes-from-universal-finite (invoked at proof line 4122), discriminant-lemma-universal-finite-syntomic-smooth (invoked at proof line 4139) and discriminant-lemma-universal-finite-syntomic-etale (invoked at proof line 4140). They allow the uniqueness statement to be checked on the universal families $V_d \to U_d$,

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whose total spaces are smooth over the base with the morphism etale over a dense open.
- Absolute Noetherian approximation. Algebra, labels algebra-lemma-colimit-category-fp-algebras (invoked at proof line 4156), algebra-lemma-colimit-lci (invoked at proof line 4161) and algebra-lemma-colimit-finite (invoked at proof line 4161). They write an affine test base as a directed limit of Noetherian affine schemes and descend the finite syntomic morphism to a stage, which is how the construction and its base change compatibility are extended beyond the locally Noetherian case.

Source reference. Original lemma 05N9: sources/stacks-chow/tags.txt, source lines 7298–7298. Navigation pointer for \ref{algebra-lemma-colimit-category-fp-algebras} used inside the counted span (line 4156); the argument that consumes it is retained in the span.

Source reference. Original lemma 07RG: sources/stacks-chow/tags.txt, source lines 9866–9866. Navigation pointer for \ref{algebra-lemma-colimit-finite} used inside the counted span (line 4162); the argument that consumes it is retained in the span.

Source reference. Original lemma 0C33: sources/stacks-chow/tags.txt, source lines 15178–15178. Navigation pointer for \ref{algebra-lemma-colimit-lci} used inside the counted span (line 4161); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FL4: sources/stacks-chow/tags.txt, source lines 19484–19484. Navigation pointer for \ref{discriminant-lemma-locally-comes-from-universal-finite} used inside the counted span (line 4122); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FL2: sources/stacks-chow/tags.txt, source lines 19482–19482. Navigation pointer for \ref{discriminant-lemma-universal-finite-syntomic-etale} used inside the counted span (line 4140); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FL1: sources/stacks-chow/tags.txt, source lines 19481–19481. Navigation pointer for \ref{discriminant-lemma-universal-finite-syntomic-smooth} used inside the counted span (line 4139); the argument that consumes it is retained in the span.

Source reference. Original proposition 0FKD: sources/stacks-chow/tags.txt, source lines 19458–19458. Navigation pointer for \ref{discriminant-proposition-tate-map} used inside the counted span (line 4068); the argument that consumes it is retained in the span.

Source reference. Original section 0BVH: sources/stacks-chow/tags.txt, source lines 14910–14910. Navigation pointer for \ref{discriminant-section-discriminant} used inside the counted span (line 4055); the argument that consumes it is retained in the span.

Source reference. Original section 0FKW: sources/stacks-chow/tags.txt, source lines 19476–19476. Navigation pointer for \ref{discriminant-section-finite-morphisms} used inside the counted span (line 4080); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FLA: sources/stacks-derham/pinned-original-source.tex, source lines 3738–3738. Navigation pointer for \ref{lemma-Garel-upstairs} used inside the counted span (line 4062); the argument that consumes it is retained in the span.

Lemma 0.1. *There exists a unique rule that to every finite syntomic morphism of schemes $f : Y \to X$ assigns $\mathcal{O}_X$-module maps*

$$\Theta^p_{Y/X} : f_*\Omega^p_{Y/\mathbf{Z}} \longrightarrow \Omega^p_{X/\mathbf{Z}}$$

satisfying the following properties

1. *the composition with $\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_X} f_*\mathcal{O}_Y \to f_*\Omega^p_{Y/\mathbf{Z}}$ is equal to $id \otimes Trace_f$ where $Trace_f : f_*\mathcal{O}_Y \to \mathcal{O}_X$ is the map from Discriminants, Section 0BVH,*
2. *the rule is compatible with base change.*

Proof. First, assume that X is locally Noetherian. By Lemma 0FLA we have a canonical map

$$c^p_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \longrightarrow f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \det(NL_{Y/X})$$

By Discriminants, Proposition 0FKD we have a canonical isomorphism

$$c_{Y/X} : \det(NL_{Y/X}) \to \omega_{Y/X}$$

mapping $\delta(NL_{Y/X})$ to $\tau_{Y/X}$. Combined these maps give

$$c^p_{Y/X} \otimes c_{Y/X} : \Omega^p_{Y/\mathbf{Z}} \longrightarrow f^*\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_Y} \omega_{Y/X}$$

By Discriminants, Section 0FKW this is the same thing as a map

$$\Theta^p_{Y/X} : f_*\Omega^p_{Y/\mathbf{Z}} \longrightarrow \Omega^p_{X/\mathbf{Z}}$$

Recall that the relationship between $c^p_{Y/X} \otimes c_{Y/X}$ and $\Theta^p_{Y/X}$ uses the evaluation map $f_*\omega_{Y/X} \to \mathcal{O}_X$ which sends $\tau_{Y/X}$ to $\mathrm{Trace}_f(1)$, see Discriminants, Section 0FKW. Hence property (1) holds. Property (2) holds for base changes by $X' \to X$ with X' locally Noetherian because both $c^p_{Y/X}$ and $c_{Y/X}$ are compatible with such base changes. For $f : Y \to X$ finite syntomic and X locally Noetherian, we will continue to denote $\Theta^p_{Y/X}$ the solution we've just found.

Uniqueness. Suppose that we have a finite syntomic morphism $f : Y \to X$ such that X is smooth over $\mathrm{Spec}(\mathbf{Z})$ and f is étale over a dense open of X. We claim that in this case $\Theta^p_{Y/X}$ is uniquely determined by property (1). Namely, consider the maps

$$\Omega^p_{X/\mathbf{Z}} \otimes_{\mathcal{O}_X} f_*\mathcal{O}_Y \to f_*\Omega^p_{Y/\mathbf{Z}} \to \Omega^p_{X/\mathbf{Z}}$$

The sheaf $\Omega^p_{X/\mathbf{Z}}$ is torsion free (by the assumed smoothness), hence it suffices to check that the restriction of $\Theta^p_{Y/X}$ is uniquely determined over the dense open over which f is étale, i.e., we may assume f is étale. However, if f is étale, then $f^*\Omega_{X/\mathbf{Z}} = \Omega_{Y/\mathbf{Z}}$ hence the first arrow in the displayed equation is an isomorphism. Since we've pinned down the composition, this guarantees uniqueness.

Let $f : Y \to X$ be a finite syntomic morphism of locally Noetherian schemes. Let $x \in X$. By Discriminants, Lemma 0FL4 we can find $d \geq 1$ and a commutative diagram

$$\begin{array}{ccccc} Y & \longleftarrow & V & \longrightarrow & V_d \\ \downarrow & & \downarrow & & \downarrow \\ X & \longleftarrow & U & \longrightarrow & U_d \end{array}$$

such that $x \in U \subset X$ is open, $V = f^{-1}(U)$ and $V = U \times_{U_d} V_d$. Thus $\Theta^p_{Y/X}|_V$ is the pullback of the map $\Theta^p_{V_d/U_d}$. However, by the discussion on uniqueness above and Discriminants, Lemmas 0FL1 and 0FL2 the map $\Theta^p_{V_d/U_d}$ is uniquely determined by the requirement (1). Hence uniqueness holds.

At this point we know that we have existence and uniqueness for all finite syntomic morphisms $Y \to X$ with X locally Noetherian. We could now give an argument

similar to the proof of Lemma 0FLA to extend to general X. However, instead it is possible to directly use absolute Noetherian approximation to finish the proof. Namely, to construct $\Theta^p_{Y/X}$ it suffices to do so Zariski locally on X (provided we also show the uniqueness). Hence we may assume X is affine (small detail omitted). Then we can write $X = \lim_{i \in I} X_i$ as the limit over a directed set I of Noetherian affine schemes. By Algebra, Lemma 05N9 we can find $0 \in I$ and a finitely presented morphism of affines $f_0 : Y_0 \to X_0$ whose base change to X is $Y \to X$. After increasing 0 we may assume $Y_0 \to X_0$ is finite and syntomic, see Algebra, Lemma 0C33 and 07RG. For $i \geq 0$ also the base change $f_i : Y_i = Y_0 \times_{X_0} X_i \to X_i$ is finite syntomic. Then

$$\Gamma(X, f_*\Omega^p_{Y/\mathbf{Z}}) = \Gamma(Y, \Omega^p_{Y/\mathbf{Z}}) = \operatorname{colim}_{i \geq 0} \Gamma(Y_i, \Omega^p_{Y_i/\mathbf{Z}}) = \operatorname{colim}_{i \geq 0} \Gamma(X_i, f_{i,*}\Omega^p_{Y_i/\mathbf{Z}})$$

Hence we can (and are forced to) define $\Theta^p_{Y/X}$ as the colimit of the maps $\Theta^p_{Y_i/X_i}$. This map is compatible with any cartesian diagram

$$\begin{array}{ccc} Y' & \longrightarrow & Y \\ \downarrow & & \downarrow \\ X' & \longrightarrow & X \end{array}$$

with X' affine as we know this for the case of Noetherian affine schemes by the arguments given above (small detail omitted; hint: if we also write $X' = \lim_{j \in J} X'_j$ then for every $i \in I$ there is a $j \in J$ and a morphism $X'_j \to X_i$ compatible with the morphism $X' \to X$). This finishes the proof. $\square$